Supplementary Material

Generalized Inverse Modeling of SIS-HOLZ Strain Profiles

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S1. Synthetic robustness check of the experimental $N = 5$ inversion

The experimental reanalysis in the main manuscript uses five surface-relaxed sine basis functions,

$$u(z) = \sum_{n=1}^{5} a_n \sin\left[n\pi\left(\frac{z}{t} + \frac{1}{2}\right)\right], \tag{1}$$

together with a fitted constant phase offset and a residual linear phase ramp. Because the representative generalized experimental solution contains two broad displacement maxima, an additional synthetic test was performed to examine whether the same $N = 5$ inversion procedure could artificially generate a double-maximum profile from a known single-maximum displacement profile.

The previously reported surface-relaxed bell-shaped displacement profile was used as the synthetic ground truth,

$$u_{\mathrm{true}}(z) = \frac{R}{2}\left[\cos\left(\frac{2\pi z}{t}\right) + 1\right], \tag{2}$$

with $R = 32.5\,\mathrm{pm}$ and $t = 166\,\mathrm{nm}$. The synthetic phase was calculated on the same diffraction-coordinate grid used for the experimental reanalysis.

Two hundred independent Gaussian phase perturbations with a standard deviation of

$$\sigma_\phi = 0.015\ \mathrm{rad} \tag{3}$$

were added to the synthetic phase profile. This value was selected to be comparable to the RMSE of the representative experimental $N = 5$ fit. It is used only as a perturbation scale and should not be interpreted as an independent estimate of the experimental measurement noise.

Each perturbed realization was fitted using the same $N = 5$ basis representation and the same nuisance parameters used in the experimental analysis. Three optimization starts were used for each realization. A recovered profile was classified as a false double-maximum solution when two or more meaningful local maxima were detected using a prominence threshold of 1 pm.

No false double-maximum recovery was observed in the 200 realizations. The median recovered peak projected displacement was 33.00 pm, with a 16th–84th percentile interval of 31.56 pm to 34.50 pm. The median phase RMSE was approximately 0.0141 rad.

These results indicate that, under random phase perturbations of the tested magnitude, the $N = 5$ inversion itself does not show a tendency to transform a single-maximum bell-shaped truth into a spurious double-maximum profile. This test does not establish that the double-maximum structure obtained from the experimental phase is uniquely physical.

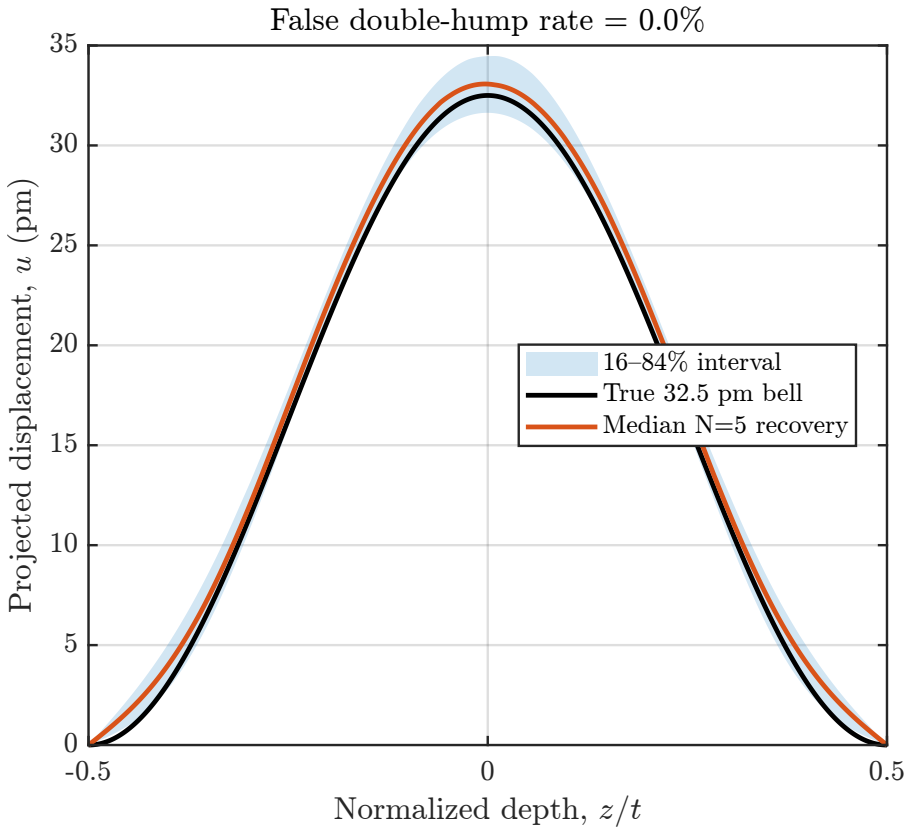


Figure 1: Synthetic robustness test of the $N = 5$ experimental inversion. The black curve is the prescribed single-maximum 32.5 pm bell-shaped displacement profile. The recovered curve represents the median of 200 independent $N = 5$ inverse solutions obtained after adding Gaussian phase perturbations with $\sigma_\phi = 0.015$ rad. The shaded region represents the 16th–84th percentile interval. No false double-maximum recovery was observed in the 200 realizations.

S2. Sensitivity of the experimental $N = 5$ solution to specimen thickness

The experimental reanalysis in the main manuscript uses the published specimen thickness of

$$t_0 = 166\,\text{nm}. \tag{4}$$

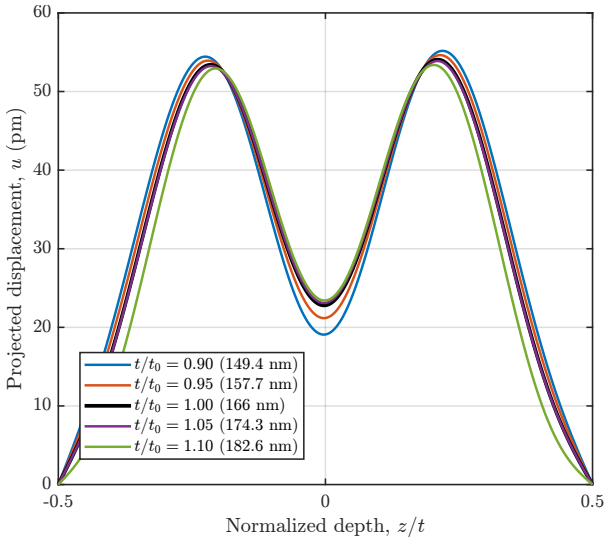
To examine whether the representative double-maximum displacement profile is generated solely by fixing the specimen thickness at this value, the complete experimental $N = 5$ inversion was repeated for assumed thicknesses

$$\frac{t}{t_0} = 0.90,\ 0.95,\ 1.00,\ 1.05,\ 1.10. \tag{5}$$

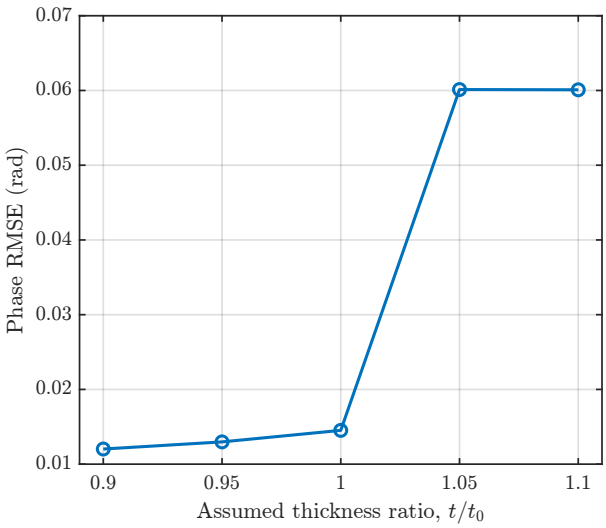
These ratios correspond to assumed specimen thicknesses of approximately

$$149.4,\ 157.7,\ 166.0,\ 174.3,\ 182.6\ \text{nm}, \tag{6}$$

respectively.



(a) Recovered $N = 5$ projected-displacement profiles for assumed thicknesses from $0.90t_0$ to $1.10t_0$. The two-maximum structure is retained throughout the tested range.



(b) Phase RMSE as a function of assumed specimen thickness. The residual increases substantially for assumed thicknesses above the published value.

Figure 2: Sensitivity of the representative experimental $N = 5$ inverse solution to the assumed specimen thickness. The reference thickness is $t_0 = 166\,\text{nm}$. At each thickness, all five displacement coefficients, the constant phase offset, and the residual linear phase ramp were refitted.

For each assumed thickness, all five displacement coefficients $a_1, \ldots, a_5$ were refitted together with the constant phase offset ϕ_0 and residual linear phase ramp c_1. All other experimental calibration and fitting conditions were retained. The recovered displacement profiles were then compared on the common normalized-depth coordinate z/t.

Figure 2 shows the resulting sensitivity analysis. The two-maximum displacement structure is retained throughout the complete tested thickness range from $0.90t_0$ to $1.10t_0$. The peak amplitudes and the depth of the central minimum change modestly with assumed thickness, but the qualitative double-maximum structure persists.

The phase RMSE remains comparatively low for assumed thicknesses from $0.90t_0$ through approximately t_0, while it increases substantially for the larger assumed thicknesses $1.05t_0$ and $1.10t_0$. Thus, within the present inverse formulation, the experimental phase is less compatible with assumed thicknesses above the published 166 nm value.

This test indicates that the representative double-maximum structure is not generated solely by fixing the specimen thickness at 166 nm. It does not, however, establish uniqueness of the recovered profile or exclude sensitivity to other instrumental or model parameters.

S3. MATLAB implementation and code availability

The MATLAB implementation associated with the main manuscript and this Supplementary Material is publicly available at:

`https://github.com/RamtinRakhsha/SIS-HOLZ-Inverse-Modeling`

The repository contains three principal folders:

- `Main_Figures/`: MATLAB scripts used for the main manuscript figures;
- `Data/`: the experimental phase dataset used for the reanalysis;
- `Supplementary/`: MATLAB scripts used for the supplementary robustness analyses.

The principal supplementary MATLAB files are:

- `Supplementary/Supplementary_False_Double_Hump_Test_N5.m`
- `Supplementary/Experimental_N5_Thickness_Sensitivity.m`

The experimental phase dataset used for the experimental reanalysis is:

`Data/ExPhase.txt`

S4. Reproducibility notes

Supplementary Fig. 1 is generated using

`Supplementary/Supplementary_False_Double_Hump_Test_N5.m`.

The script uses a fixed random-number seed and, in the calculation reported here, 200 independent phase perturbations.

Supplementary Fig. 2 is generated using

`Supplementary/Experimental_N5_Thickness_Sensitivity.m`.

This script repeats the full $N = 5$ experimental inversion at five assumed specimen thicknesses while refitting the five displacement coefficients and the two nuisance phase parameters at each thickness.

The supplementary analyses are intended to evaluate specific robustness questions associated with the representative experimental $N = 5$ solution. They do not establish a unique through-thickness displacement field. Their purpose is to determine whether the observed double-maximum structure can be explained simply by random phase perturbations or by fixing the specimen thickness at its published value.